DEALER TRAINING · NOVEMBER 2026

Cala-Mag V



Calcium, magnesium, boron and nitrogen in 1 acidic liquid: N 10, Ca 6, Mg 4 and B 1 grams in every 100 ml.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

Cala-Mag V supplies calcium, magnesium and boron in 1 bottle, with nitrogen: N 10, Ca 6, Mg 4 and B 1, in grams in every 100 ml. It is acidic so that the calcium stays dissolved. Pakistani soils hold a lot of calcium as lime, but the plant can still be short of calcium, because the root cannot take up the calcium in lime. A heavy potash dose also reduces calcium uptake at the root. Calcium does not move inside the plant, so the fruit, the tuber and the growing tip show the shortage first: blossom end rot on tomato, tip burn on leafy crops and hollow heart in potato. It has to be applied while the fruit or tuber forms, and more than once. The crop plans name a spray for tomato at early and grand growth and for chili at grand growth; potato gets 4 bottles through the water at germination and 4 at grand growth; strawberry, mango and guava take it through the water.

The 4 points it must cover

1. The bottle: 1 L, N 10, Ca 6, Mg 4, with boron, PS 5469-2020. Every litre holds 60 g of calcium and 40 g of magnesium. Boron is 1% of the product; the label names it without a figure.
2. Even in soil rich in lime, the plant can be short of calcium. A heavy potash dose makes the shortage worse.
3. Timing: calcium does not move inside the plant, so it must be applied while the fruit or tuber forms, at the stages the crop plan names, usually more than once.
4. Method: spray on tomato and chili; through the water on potato, strawberry, mango and guava. The crop plan row gives the method.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the bottle, in the published crop plan or in the named evidence on the slides that follow.

THE BOTTLE

What the bottle says, and what it is registered as.

1 L

the bottle. Every litre holds 100 g of N, 60 g of Ca and 40 g of Mg by the declared analysis, and 10 g of B by VAN's published analysis.

Ca 6 · Mg 4

weight in volume (w/v): 6 g of calcium and 4 g of magnesium in every 100 ml. VAN's product data says the ratio is set so that neither blocks the uptake of the other.

N 10

nitrogen, 100 g per litre, to support growth while the fruit forms.

B 1

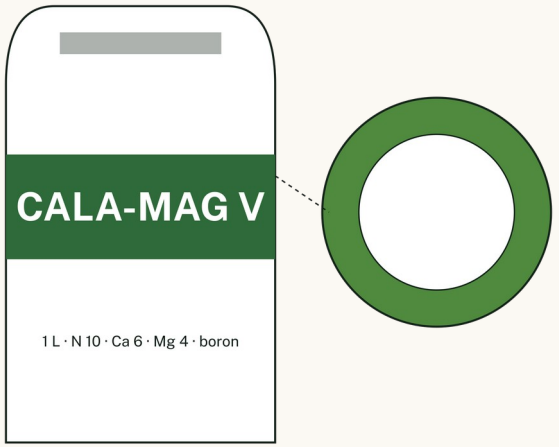
boron, 10 g per litre, as VAN publishes the analysis; the label names boron without a figure. It moves with the calcium to the growing point.

pH 2 to 4

acidic, transparent and 100% soluble, as VAN's product data gives it. Calcium comes out of solution as the pH approaches neutral, so the acid keeps it dissolved.

2 routes

fertigation and foliar, as VAN names them. The crop plan row gives the method; the guava row gives a spray rate of 500 ml per 100 L of water.



Registered as	Pakistan Standard	Manufacturing licence
Cala-Mag V	PS 5469-2020	CM/L-4299/2026

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“My tomatoes came out with a black sunken patch at the bottom. The spray man said it was a disease. 3 sprays later it was still there. What is it?”

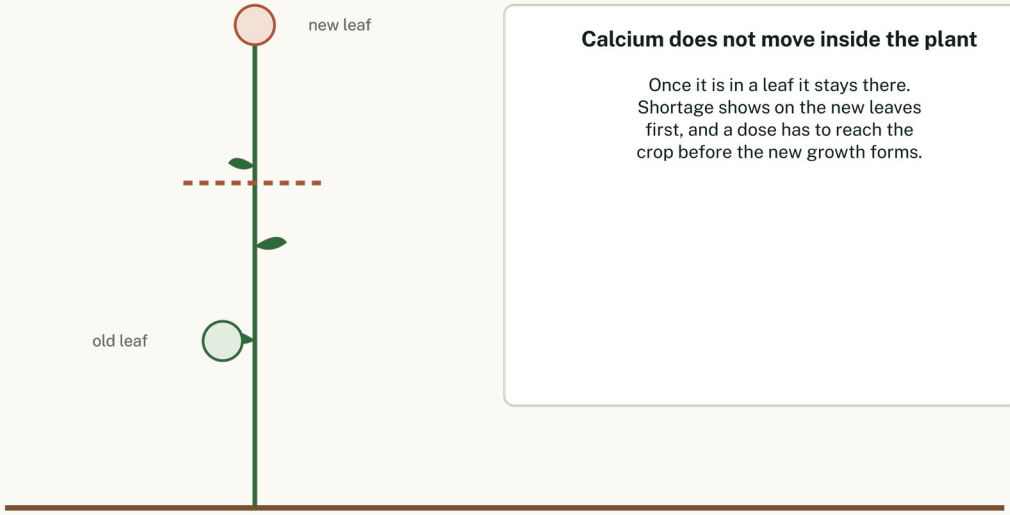
The fruit did not get enough calcium while it was forming, and no disease spray can supply calcium. VAN developed this product after a similar case in Sindh: a tomato crop with patches on the fruit was being treated for a virus, and VAN’s team found a calcium shortage on soil rich in calcium. The answer has 3 parts: why fruit can be short of calcium in lime-rich soil, why calcium applied late cannot reach the fruit, and how the bottle keeps the 3 nutrients dissolved.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

A plant can be short of calcium even in soil rich in lime.

Calcium builds the cell wall and is needed at the tip of every growing point. In the Punjab soil survey (770,160 samples, tested in about 2016 to 2018), the average calcium carbonate (lime) content was 6.53%. The root cannot take up the calcium in lime; it takes up the free calcium dissolved in the soil water, and a heavy potash or magnesium dose reduces that uptake at the root. Inside the plant, calcium does not move once it is in place. So the oldest leaves look healthy while the youngest tissue is damaged: blossom end rot on tomato, tip burn on leafy crops and hollow heart in potato. Magnesium does move inside the plant, so a magnesium shortage shows first as yellowing between the veins of the old leaves.



Calcium does not move inside the plant

Once it is in a leaf it stays there. Shortage shows on the new leaves first, and a dose has to reach the crop before the new growth forms.

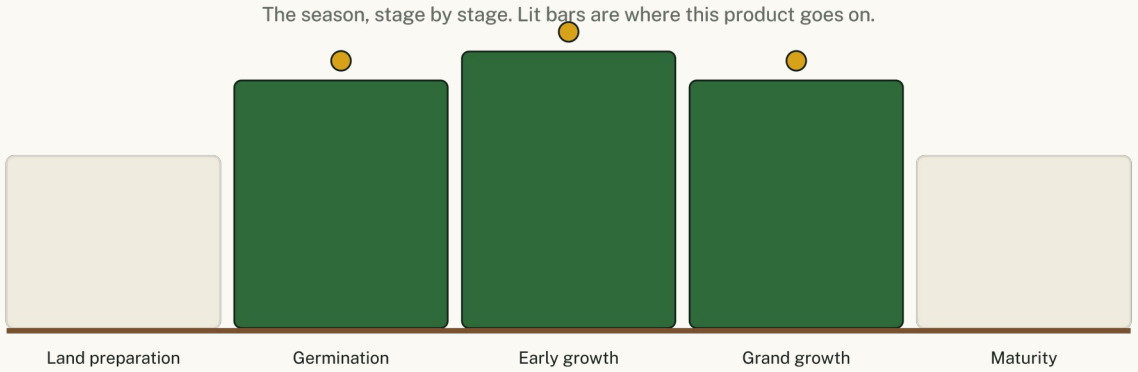
Say it like this: “The calcium in your soil is lime, and the root cannot take it up. Calcium in lime and calcium the plant can use are 2 different things.”

Calcium the fruit did not receive while it was forming cannot reach it later. This is established plant science, not a VAN measurement.

ANSWER 2 OF 3

Calcium has to be applied while the fruit is forming.

Calcium does not move inside the plant once it is in place, so the fruit and the tuber take it up only while they are forming. Calcium applied after the fruit has formed cannot move into it. Each crop plan names its own stages, and most repeat the dose. Tomato: early growth and grand growth. Chili: grand growth. Potato: germination and grand growth. Strawberry: early growth and grand growth. Mango: grand growth. Guava: September, flowering in October and fruiting in November, and the October and November doses may each be repeated 15 days later. VAN advises several small applications rather than 1 large one, because of how calcium behaves in the plant.



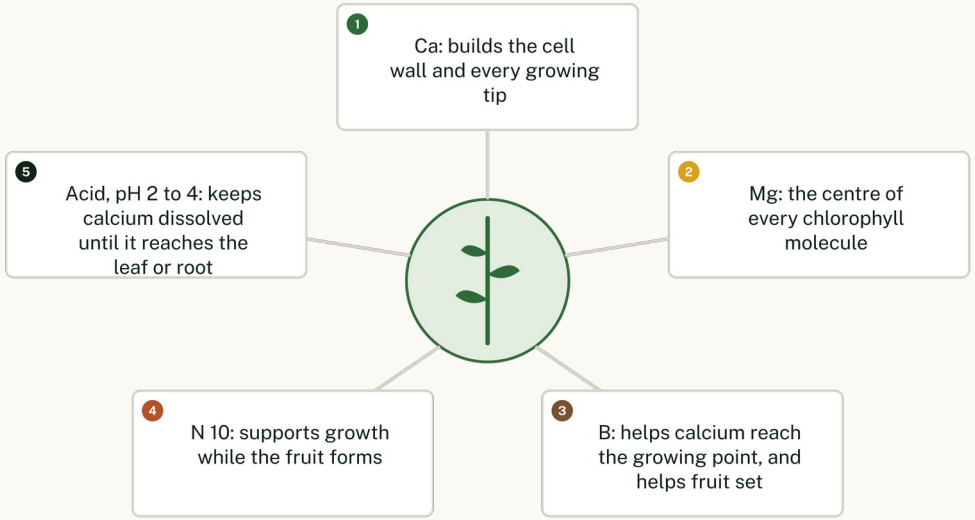
The bars show the stages named in the crop plans: potato from germination; tomato and strawberry from early growth; chili and mango at grand growth. The rows are on the crop plan slides.

Say it like this: “Apply it while the fruit is forming, and apply it more than once. Calcium applied after the fruit has formed cannot get into it.”

ANSWER 3 OF 3

Calcium, magnesium and boron are kept dissolved together in 1 bottle.

Calcium and magnesium compete for the same uptake sites, so in a badly balanced mix each can block the uptake of the other. VAN's product data says the ratio in this bottle is set so that neither blocks the other. Boron is needed for calcium to reach the growing point and to build the cell wall, and for flowers and fruit to set. The liquid is acidic, pH 2 to 4, because calcium comes out of solution as the pH approaches neutral; in alkaline, lime-rich soil, dissolved calcium also comes out of solution. Amino acids are in the formulation to help the 3 nutrients pass through the leaf surface instead of staying on it. The nitrogen supports growth while the fruit is forming.



The 5 parts of the formulation, from VAN's product data. The label gives no figure for the amino acids.

Say it like this: "Calcium and magnesium can block each other. In this bottle the ratio is set so that neither blocks the other, and the acid keeps all 3 dissolved."

THE EVIDENCE

What is verified for Cala-Mag V, and what is not.

No trial of Cala-Mag V is on VAN's record. A dealer may state the following as facts, and should say where each comes from: what the bottle declares, the licence, the crop plan rows, the Punjab soil survey and the chemistry.

Item	What it says	How a dealer may use it
The bottle	Cala-Mag V, N 10%, Ca 6%, Mg 4% w/v, with boron (1% in the product, as VAN publishes it; the label names boron without a figure); transparent liquid, pH 2 to 4, 100% soluble, amino acids in the formulation; 1 L.	As fact. Read N, Ca and Mg off the bottle in hand; the boron figure is VAN's published analysis.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-4299/2026.	As fact. The number, never an expiry date.
The crop plan rows	11 rows on 6 crop plans. Spray: chili at grand growth, tomato at early and grand growth. Fertigation: potato 4 bottles at germination and 4 at grand growth; strawberry 4 at early growth and 1 at grand growth; mango 1 at grand growth; guava 1 in September (or spray at 500 ml per 100 L), then 1 at flowering and 1 at fruiting, each of these 2 may be repeated 15 days later.	As the published rate, exactly as printed.
The Punjab soil survey (770,160 samples, about 2016 to 2018)	Mean calcium carbonate 6.53%; 69.1% of samples above pH 8; 51.7% low in boron; 87.8% below 180 parts per million (ppm) available potassium, the level Punjab soil laboratories treat as adequate.	As fact about the soil: high in lime, often low in boron. Never as product performance.
The chemistry	Calcium is immobile in the plant; magnesium is mobile. Calcium and magnesium compete at the root; a heavy dose of potash suppresses both. Calcium comes out of solution as a solution approaches neutral pH.	As general plant nutrition, said as such.
Not on VAN's record	A Cala-Mag V trial; a yield, grade, firmness or shelf-life figure; a blossom-end-rot cure rate; an amino acid figure; a compatibility chart; a Pakistani figure for how much of the country's soil is calcareous.	Not said.

Sources: the bottle; the Pakistan Standard and licence register; the crop plans on van.com.pk; the VAN farmer price sheet, Aug 2026; the Punjab soil survey (770,160 samples, tested in about 2016 to 2018); VAN's 17-nutrients page.

On tomato, 2 bottles per acre cost Rs 1,980 and put 120 g of calcium and 80 g of magnesium on the crop.

The price is the delivered farmer price from the VAN farmer price sheet, Aug 2026. Nutrients are from the declared analysis, w/v: 6% is 60 g in a litre.

Cala-Mag V, N 10 · Ca 6 · Mg 4 · B 1	Volume	N	Ca	Mg	B	Rupees
1 bottle	1 L	100 g	60 g	40 g	10 g	990
Tomato, 2 bottles, sprayed	**2 L**	**200 g**	**120 g**	**80 g**	**20 g**	**1,980**
Potato, 8 bottles, through the water	**8 L**	**800 g**	**480 g**	**320 g**	**80 g**	**7,920**

On tomato, 2 bottles per acre, sprayed at early growth and grand growth, cost Rs 1,980 a season and put 120 g of calcium and 80 g of magnesium (declared analysis) and 20 g of boron (VAN's published analysis) on the crop. On potato, 8 bottles per acre through the water cost Rs 7,920 and supply 480 g of calcium to the root zone around the tubers. VAN publishes no figure for how much of this the crop takes up.

Cala-Mag V 1 L at Rs 990: the VAN farmer price sheet, Aug 2026. 60 g = 1,000 ml × 0.06; 10 g = 1,000 × 0.01. Tomato 2 rows × 1 bottle; potato 2 rows × 4 bottles (the crop plans on van.com.pk). VAN's potato cost plan gives the same 8 bottles at Rs 990 each.

Per acre, the crop plans use 1 bottle on chili and mango and up to 8 bottles on potato.

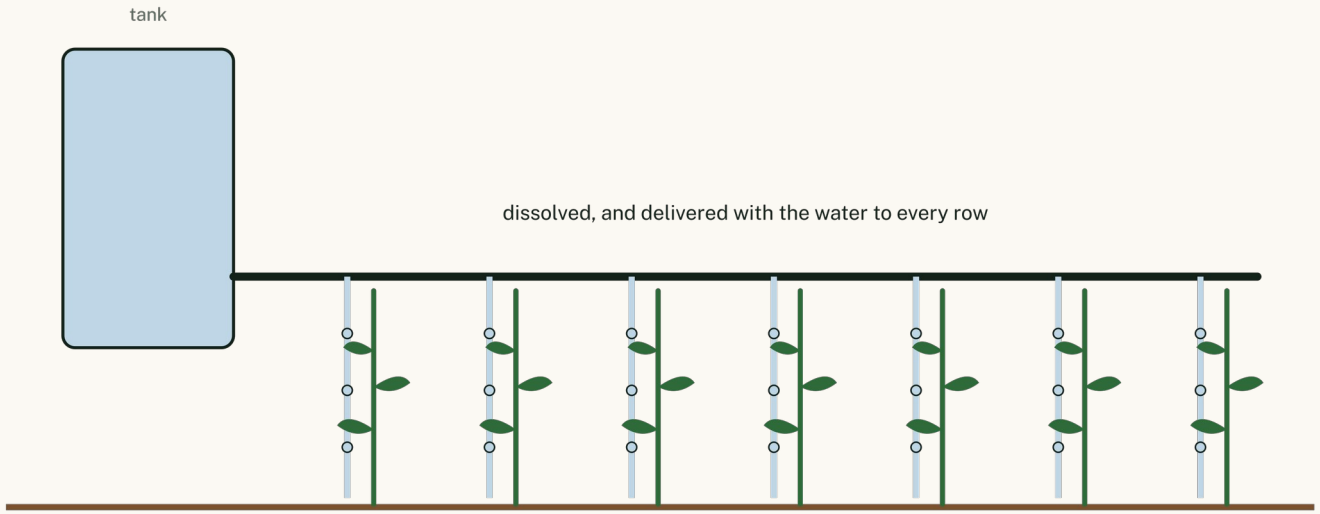
Bottles per acre are from the published crop plans. Rupees per acre are calculated from the delivered farmer price in the VAN farmer price sheet, Aug 2026. The stage and method are copied from the crop plan.

Crop	Stage and method, as printed	Bottles	Rupees	Ca per acre
Chili	Grand growth, spray	1	990	60 g
Tomato	Early growth, spray; grand growth, spray	2	1,980	120 g
Potato	Germination, 4 bottles, fertigation; grand growth, 4 bottles, fertigation	8	7,920	480 g
Strawberry	Early growth, 4 bottles, fertigation; grand growth, 1 bottle, fertigation	5	4,950	300 g
Mango	Grand growth, fertigation	1	990	60 g
Guava	September, fertigation or spray at 500 ml per 100 L; October flowering and November fruiting, fertigation, each of these 2 may be repeated 15 days later	3	2,970	180 g

Potato uses the most bottles, because the tuber forms underground over several weeks and calcium reaches it only from the soil water, so it is applied through the irrigation water. Guava is the only crop whose row gives a spray rate: 500 ml per 100 L of water.

Rows from the crop plans on van.com.pk, as printed; 1 L at Rs 990. Guava: 3 rows. The flowering row adds “repeat 15 days after flowering” and the fruiting row adds “repeat 15 days after fruiting”. The repeats are optional, so this deck counts 3 bottles per acre; the crop plan prints no total. Ca per bottle 60 g.

Apply it through the water or as a spray, more than once while the fruit forms, and keep the liquid acidic.



Fertigation: dissolve, then release with the water flowing

For potato, strawberry, mango and guava. Stir it into the drum or drip tank and release it steadily with the irrigation water. Add nothing that raises its pH, and keep phosphate and sulfate fertilisers out of the tank: calcium comes out of solution with either. For any other mix, do a jar test: mix the products in a jar of water at the tank ratio, and do not use the mix if anything settles.

Spray: water first, product second, morning or evening

For the chili and tomato rows. Fill the tank with clean water, add the product and keep the tank agitated. On tomato and chili, put 1 bottle (1 L) per acre in 100 to 200 L of water. The guava row prints its own rate: 500 ml per 100 L. Use a fine, even fan to cover the leaves and young fruit without run-off. Do not spray at midday.

While the fruit forms, more than once

Apply it at the stages the crop plan names: early and grand growth on tomato, germination and grand growth on potato, and the optional 15-day repeats on guava. Calcium applied after the fruit has formed cannot move into it, and the fruit forms over all of these stages, so several applications work better than 1 large one. Do not exceed the crop plan rate: every bottle contains boron, and the margin between enough boron and too much is narrow.

OBJECTION 1 OF 7

“My soil is full of calcium. Why would I buy calcium?”

What the dealer says

Because the calcium in your soil is in lime, and the root cannot take up calcium from lime. It takes up the free calcium in the soil water, and a heavy potash dose reduces that uptake at the root. Inside the plant calcium cannot move, so the fruit and the tuber, which form last, are the first to run short. This bottle puts calcium, with magnesium and boron, where the fruit is forming, at the time it is forming. On tomato it is 1 bottle at Rs 990 at early growth and another at grand growth: 2 bottles, Rs 1,980 per acre.

The facts behind the answer

- Calcium in lime and calcium the root can take up are 2 different things (VAN’s 17-nutrients page).
- Mean calcium carbonate 6.53% across 770,160 samples (the Punjab soil survey, about 2016 to 2018).
- Cala-Mag V 1 L at Rs 990 (VAN farmer price sheet, Aug 2026); tomato 2 bottles, early and grand growth, spray (the crop plans on van.com.pk).

Never say: “It gives you X more maunds”, “firmer fruit by X%”, or a rupees-per-acre return. No trial exists.

OBJECTION 2 OF 7

“Can I put it in the tank with my potash or my phosphate?”

What the dealer says

Not with a phosphate or a sulfate, and sulfate of potash (SOP) is a sulfate: calcium comes out of solution with either, so you lose the calcium and may block the drip line. This happens with any calcium product; it is basic chemistry. Apply it on its own, or do a jar test first if you must mix it with a potash other than SOP. Also keep it acidic: the liquid in the bottle is pH 2 to 4 because calcium comes out of solution as the pH approaches neutral, so do not add anything that neutralises it.

The facts behind the answer

- VAN’s guidance: the acidity keeps the calcium dissolved; do not tank-mix it with anything that raises the pH.
- Calcium phosphate and calcium sulfate are insoluble; they settle out in a tank (textbook chemistry).
- No compatibility chart for Cala-Mag V is published. Jar test: mix the products in a jar of water at the tank ratio, and do not use the mix if anything settles.

Never say “Mix it with anything”, and never give a compatibility list VAN has not published.

OBJECTION 3 OF 7

“Which stage, and how many bottles?”

What the dealer says

Use what the plan prints. Tomato: 1 bottle sprayed at early growth and 1 at grand growth. Chili: 1 at grand growth. Potato: 4 bottles through the water at germination and 4 again at grand growth. Strawberry: 4 at early growth and 1 at grand growth, through the water. Mango: 1 at grand growth. Guava: 1 in September, then 1 at flowering in October and 1 at fruiting in November; each of these 2 may be repeated 15 days later. Do not apply it once the fruit has formed: by then the calcium cannot get into the fruit.

The facts behind the answer

- Tomato 2 × 1 bottle, spray; chili 1 bottle, spray; mango 1 bottle, fertigation (the crop plans on van.com.pk).
- Potato 4 + 4 bottles, fertigation; strawberry 4 + 1 bottles, fertigation; guava 3 rows, 2 of them with an optional 15-day repeat (the crop plans on van.com.pk).
- VAN advises repeated small doses instead of a single large dose (VAN guidance).

Never say: “1 big dose when you see the rot”, or a rate for cotton or any orchard crop other than mango and guava; they have no crop plan row.

OBJECTION 4 OF 7

“Spray or through the water? Which is better?”

What the dealer says

The crop plan row for your crop says which, and there is a reason for each. Tomato and chili are sprayed onto the leaf and the young fruit, 1 bottle per acre in 100 to 200 L of water, because the calcium is needed there and cannot move there from elsewhere in the plant. Potato takes it through the water because the tuber forms underground. Strawberry, mango and guava also take it through the water, because their crop plan rows say fertigation. For guava the row gives both: fertigation, or a spray at 500 ml per 100 L of water.

The facts behind the answer

- VAN names fertigation and foliar as the routes for Cala-Mag V.
- Spray on chili and tomato; fertigation on potato, strawberry, mango and guava; both on the guava September row (the crop plans on van.com.pk).
- Calcium is immobile in the plant, so it has to be supplied while the fruit is forming (VAN’s 17-nutrients page).

Never say or give: a spray rate for a fertigated crop, or a fertigation rate for a sprayed crop, unless the crop plan prints it.

OBJECTION 5 OF 7

“The rot is already on the fruit. Will this cure it?”

What the dealer says

No product puts calcium into a fruit that has already formed, and VAN does not say this one does. The fruit with the patch is lost. The spray reaches the flowers and young fruit that are forming now, at the stage the plan names, so the next fruit forms with calcium in its cell walls. Apply it again at the next stage the plan names.

The facts behind the answer

- Calcium does not move inside the plant once placed; the youngest tissue fails first (VAN’s 17-nutrients page).
- Calcium arriving after the fruit has formed cannot move into it (VAN guidance; textbook plant nutrition).
- VAN has no figure showing that it cures blossom end rot.

Never say: “It cures blossom end rot” or “it stops the rot in a week”. No cure rate exists.

OBJECTION 6 OF 7

“There is boron in it. Will it burn my crop?”

What the dealer says

Not at the crop plan rate. The crop needs boron in grams per acre, and the margin between enough and too much is narrow. That is why the boron is in this bottle at a fixed amount, instead of being applied separately. 1 bottle contains 10 g of boron; the tomato crop plan puts 20 g on the acre over the season. Use the number of bottles the plan gives and no more, because extra bottles add boron the crop does not need.

The facts behind the answer

- The narrow margin between enough and too much boron (VAN’s 17-nutrients page).
- B 1% = 10 g per litre; tomato 2 bottles = 20 g; potato 8 bottles = 80 g (VAN’s published analysis and the crop plans on van.com.pk).
- 51.7% of samples tested low in boron (the Punjab soil survey, about 2016 to 2018).

Never say: “Boron cannot harm a crop” or “double the bottles on a boron-short field”.

OBJECTION 7 OF 7

“Is it registered? What is the licence?”

What the dealer says

It is registered under Pakistan Standard PS 5469-2020 for liquid nutrient supplements, with manufacturing licence CM/L-4299/2026, and VAN is the registered maker. Every batch is tested before it ships at VAN Lab, which the Pakistan National Accreditation Council (PNAC) accredits as LAB 336. Enter the batch number from the bottle on van.com.pk, and you can read the certificate for that batch.

The facts behind the answer

- Cala-Mag V: PS 5469-2020, CM/L-4299/2026 (the licence register on van.com.pk).
- Every batch is tested at VAN Lab, PNAC LAB 336, before it ships.
- A grower can read the certificate for any batch on van.com.pk by its batch number.

Never say or give: a licence expiry date, or an amino acid figure as if it were on the certificate.

Cala-Mag V in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Potato	Germination	4 packs	Fertigation	1 L
Potato	Grand Growth	4 packs	Fertigation	1 L
Tomato	Early Growth	1 pack	Spray	1 L
Tomato	Grand Growth	1 pack	Spray	1 L
Chili	Grand Growth	1 pack	Spray	1 L
Mango	Grand Growth	1 pack	Fertigation	1 L

The rows are printed as published in the crop plans on van.com.pk. Coverage: these are all 11 rows in the 6 crop plans that name Cala-Mag V. Of these 11 rows, 7 say fertigation, 3 say spray and 1 says fertigation or spray. Every row uses the 1 L pack. Each rate is per acre, and the crop page scales it to the size of the field.

Cala-Mag V in the published crop plans, continued.

Crop	Stage	Rate per acre	Method	Pack
Strawberry	Early Growth	4 packs	Fertigation	1 L
Strawberry	Grand Growth	1 pack	Fertigation	1 L
Guava	September	1 pack	Fertigation, or spray at 500 ml per 100 L water	1 L
Guava	October, flowering starts	1 pack	Fertigation. Repeat 15 days after flowering	1 L
Guava	November, fruiting	1 pack	Fertigation. Repeat 15 days after fruiting	1 L

The rows continue from the previous slide. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A yield, grade, firmness or shelf-life figure. No Cala-Mag V trial exists, so the effect of calcium on fruit grade is described without a number.

2. That it cures blossom end rot, tip burn or hollow heart, or any disease. Calcium cannot enter a fruit that has already formed.

3. An amino acid figure, or any figure the bottle does not declare, as if it were on the certificate.

4. A rate for cotton or for any orchard crop other than mango and guava. They have no crop plan row.

5. That it corrects a soil that is short of calcium, or changes the soil's calcium or pH. It feeds the plant while the fruit forms; it does not change the soil, so a soil test will not show a difference.

6. That it mixes with other products. It must not go with a phosphate or sulfate, other mixes need a jar test, and VAN has published no compatibility chart.

HOW A GROWER VERIFIES A BOTTLE

Every batch is tested, and the batch number on the bottle identifies its certificate.

1 · The batch number

Every bottle carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any bottle by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for Cala-Mag V shows

The certificate gives the tested values for the batch against the declared analysis: nitrogen, calcium and magnesium against N 10, Ca 6, Mg 4 w/v, and boron where the certificate includes it. It also shows the batch number, and the Pakistan Standard PS 5469-2020 and licence CM/L-4299/2026 printed on the bottle. The amino acids have no label figure and are not on the certificate. This training shows no batch result; a grower can read the certificate of his own batch on van.com.pk by its batch number.

THE URDU LINES · REVIEW STATUS UNDER EACH LINE

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

Calcium is needed while the fruit forms, not later. *English meaning · Urdu reviewed 27 Sep 2026*	کیلشیم پھل بنتے وقت چاہیے، بعد میں نہیں
The soil has plenty of calcium, yet the plant is short of calcium, because it is not in a form the plant can take up. *English meaning · Urdu line pending review*	مٹی چونے سے بھری ہے، پھر بھی پودے کو کیلشیم نہیں ملتا
At the stages the crop plan names, a little at a time, more than once. *English meaning · Urdu line pending review*	پھول سے پھل بھرنے تک، تھوڑا تھوڑا، بار بار
Heavy potash blocks calcium and magnesium at the root. *English meaning · Urdu reviewed 27 Sep 2026*	زیادہ پوٹاش جڑ پر کیلشیم اور میگنیشیم کو روک دیتی ہے

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

Cala-Mag V supplies calcium, magnesium and boron in 1 bottle, applied while the fruit or tuber forms, usually more than once.

